

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)
FACULTY OF MANAGEMENT STUDIES (FMS)
BBA PROGRAMME

SEMESTER – II –UNIVERSITY EXAMINATION, APRIL 2018

BM110 BUSINESS STATISTICS (OLD)

BM110.1 BUSINESS STATISTICS (NEW)

Date: 07/05/2018, Monday

Time: 10.00 AM to 10.30 AM

Total Marks: 15

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Important Instructions:

- All questions are compulsory.
- This is a closed book examination.
- There are two sections in the question paper. Section – I contains Q. No.: 1 which is of 15 marks and Section – II contains total five questions wherein Q. No.: 2 to 5 are of 10 marks each and Q. 6 is a Case study of 15 marks.
- Section – I should be answered in the given question paper and Section – II should be answered in a separate answer book.
- Maximum time allotted for Section – I is 30 minutes. Answer Book for Section – II will be given when the Question Paper for Section – I is returned. However, candidates who complete Section – I earlier than 30 minutes, can collect Answer Book for Section – II immediately after handing over Section – I question paper.
- Figures on the right indicate marks for each question.

SECTION – I

Q.1 Objective Type Questions:

A) Match the following: (i – vi)

15
06

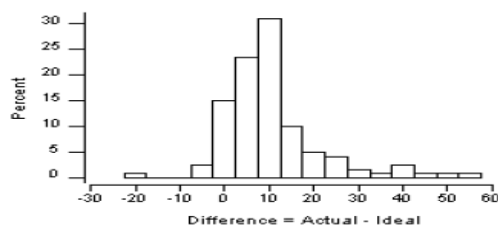
Q. No.:		Probable Answer Keys		Write Your Answer Here
i.	Convenience Sampling	a)	Measures the effect of treatment	i. -
ii.	Systematic Sampling	b)	Used for prediction	ii. -
iii.	Chi-square test	c)	Measure the difference in mean of two groups	iii. -
iv.	Correlation	d)	Hypothesis testing of association	iv. -
v.	Regression	e)	Null Hypothesis	v. -
vi.	Paired sample t-test	f)	Alternate Hypothesis	vi. -
		g)	Non-probability Sampling	
		h)	Probability Sampling	
		i)	Hypothesis testing of relation	
		j)	Known population mean	

B) Multiple-Choice Questions (vii – ix) (Note: Encircle the correct answer)**03**

vii. Which of the following would indicate that a dataset is not bell-shaped?

- a) The range is equal to 5 standard deviations.
- b) The range is larger than the interquartile range.
- c) The mean is much smaller than the median.
- d) There are no outliers.

viii. The following histogram shows the distribution of the difference between the actual and “ideal” weights for 119 female students. What is the approximate shape of the distribution?



- a) Nearly symmetric
- b) Skewed to the left
- c) Bimodal
- d) Skewed to the right

ix. Which formula is used to calculate degrees of freedom of the chi-square test for a contingency table with “r” rows and “c” columns?

- a) $r \cdot c$
- b) $r \cdot (c - 1)$
- c) $(r + 1) \cdot (c + 1)$
- d) $(r - 1) \cdot (c - 1)$

C) Fill-in the Blanks (x – xii) You may choose from the following answers:**03**

Accept a false null hypothesis	Reject true null hypothesis	Mutually exclusive	Collectively exhaustive	Two	More than two
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x. Type II error associated with hypothesis is committed when _____.

xi. Events are _____ if they do not share any common outcome of a random experiment.

xii. Analysis of Variance (ANOVA) is used when nominal data has _____ groups.

D) True or False (xiii – xv) (Note: Encircle the correct answer)**03**

xiii. One or two tail test will determine if the region of rejection is located in one or two tails of the distribution. (True) or (False)

xiv. By taking a level of significance of 5% it is the same as saying we are 95% confident that the results have not occurred by chance. (True) or (False)

xv. Educational qualification is an ordinal level of data. (True) or (False)

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BM110 BUSINESS STATISTICS (OLD)
BM110.1 BUSINESS STATISTICS (NEW)

Date: 07/05/2018, Monday

Time: 10.30 AM to 01.00 PM

Total Marks: 55

Note: - (i) All questions are compulsory.

(ii) This is a closed book examination.

(iii) There are two sections in the question paper. **Section – I** contains Q. No.: 1 of 15 marks and **Section – II** contains total five questions where in Q. No.: 2 to 5 are of 10 marks each and Q. 6 is a Case study of 15 marks.

(iv) **Section – I** should be answered in the given question paper and **Section – II** should be answered in a separate answer book.

(v) Maximum time allotted for **Section – II** is 150 minutes.

(vi) Figures on the right indicate marks for each question.

SECTION – II

Q.2 Identify the level of measurement for each of the scenario given below: **10**

- a. The teacher of a class of third graders records the height of each student.
- b. The teacher of a class of third graders records the eye color of each student.
- c. The teacher of a class of third records the grade for mathematics for each student.
- d. The teacher of a class of third graders records the percentage that each student got correct on the last science test.
- e. A scientist makes a list of temperatures in degrees Celsius for the month of May.
- f. A film critic lists the top 50 greatest movies of all time.
- g. The roster of a basketball team lists the jersey numbers for each of the players.
- h. A local animal house keeps track of the breeds of dogs that come in.
- i. A local animal house keeps track of the weights of dogs that come in.
- j. A sports trainer makes a list of left handed bowler in a school.

OR

Q.2 Describe the following sampling plans with examples and give their relative merits and demerits. **10**

- a. Judgment Sampling
- b. Stratified Sampling
- c. Systematic Sampling
- d. Cluster Sampling
- e. Quota Sampling

Q.3 The table mentioned shows that 10 highest-paid chief executive officers of the last decade. **10**

- a. Calculate the mean, median, mode and range for the 10 highest-paid chief executive officers. When can mean or median can be used as measure of central tendency?
- b. Does the mean accurately reflect the center of the data?
- c. Calculate standard deviation and variance for the above data. Interpret it.

Name	Firm	Compensation (in \$ millions)
Ellison	Oracle	1,835.7
Diller	Expedia	1,142.9
Irani	Occidental Petroleum	857.1
Jobs	Apple	748.8
Richard	Capital One	568.5
Mozilo	Countrywide	528.6
Isenberg	Nabors Industries	518.0
Semel	Yahoo	489.6
Henry	Cedant	481.2
William	UnitedHealth Group	469.3

Source: The Wall Street Journal, July 27, 2010

- Q.4** A research study was conducted to examine the differences between older and younger adults on satisfaction towards ruling BJP government. A pilot study was conducted to examine this hypothesis. Ten older adults (over the age of 70) and ten younger adults (between 20 and 30) were given a questionnaire which has statements related to satisfaction towards government and measured on 5 point Likert scale. Scores range from 0 to 60 with high scores means high satisfaction; low scores means low satisfaction. The data are presented below. **10**

Older Adults	Younger Adults
45	34
38	22
52	15
48	27
25	37
39	41
51	24
46	19
55	26
46	36

- Suggest a suitable test to be performed on given data set.
- Mention the assumptions and level of data required for the test.
- What is your calculated answer?
- What would be the null hypothesis in this study?
- What would be the alternate hypothesis?
- What significance level did you choose and why?
- What is your tabulated value?
- Interpret your answer.

OR

Q.4 *Frame Null and Alternate Hypotheses for the below mentioned cases:* **10**

- a. A trade group predicts that back-to-school spending will average \$606.40 per family this year. A different economic model is needed if the prediction is wrong. Specify the null and alternate hypotheses to determine if a different economic model may be needed.
- b. An advertisement for a popular weight-loss clinic suggests that participants in its new diet program lose, on an average, more than 10 pounds. A customer activist wants to determine if the advertisement's claim is valid. Specify the null and the alternative hypotheses to validate the advertisement's claim.
- c. A television research analyst wishes to test a claim that more than 50% of the households will tune in for a TV episode. Specify the null and the alternative hypotheses to test the claim.
- d. It is generally believed that at least 0.60 of the residents in a small town in Texas are happy with their lives. A sociologist is concerned about the lingering economic crisis and wants to determine whether the crisis has adversely affected the happiness level in this town. Specify the null and alternative hypotheses to determine if the sociologist's concern is valid.

Q.5 Define the following terms with suitable examples: **10**

- a. Sample space
- b. Event
- c. Probability
- d. Significance level
- e. Rejection region
- f. Normal distribution
- g. One sample t-test
- h. Degrees of freedom
- i. Goodness of fit
- j. Test of independence

Q.6 **Case Study / Case-let:** **15**

The time “x” in years that an employee spent at a company and the employee’s hourly pay “y”, for 5 employees are listed in the table below. Calculate and interpret the correlation coefficient r. Find r^2 , the equation of the regression line. Interpret r^2 . Test the hypothesis at 5% significance level for both correlation and regression. Mention the objectives, assumptions and hypothesis of correlation and regression.

No. of Years (x)	Employees Hourly Wage (y)
5	25
3	20
4	21
10	35
15	38